

Seminar: Image Segmentation Based on Visual Prompt

Group MNIST

VNUHCM - University of Science

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General Background

Image segmentation is a fundamental task in computer vision that involves partitioning a digital image into multiple segments or sets of pixels, often corresponding to different objects or parts of objects.



(a) Image



(b) Semantic segmentation



(c) Instance segmentation



(d) Panoptic segmentation

The history of this field can be broadly divided into three distinct eras.

- **Pre-Deep Learning Era (c. 1970s–2000s)**
- **Deep Learning Era (c. 2010s)**
- **Modern Research Directions**



Pre-Deep Learning Era (c. 1970s-2000s)

Segmentation methods were primarily based on handcrafted features and mathematical models. These techniques required significant domain knowledge and focused on pixel-level properties such as intensity, color, texture, and edges.



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These methods, while foundational, relied on heuristics and low-level features, often struggling with complex scenes and semantic understanding.



Deep Learning Era (c. 2010s)

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Key milestones include:

- Fully Convolutional Networks (FCN)
- U-Net
- Mask R-CNN



Current research continues to build upon these deep learning foundations, exploring new architectures and learning paradigms for greater accuracy, efficiency, and generalization.

- Transformer-Based Architectures
- Foundation Models
- Self-Supervised Learning



The drive to advance image segmentation stems from both scientific inquiry and practical necessity.

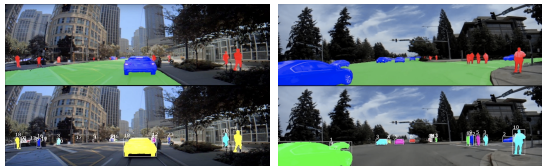
Scientifically, segmentation addresses a fundamental question in perception: while image classification tells us what is in an image, segmentation answers where it is, down to the pixel level.

Practically, segmentation is an enabling technology for a multitude of real-world applications.



Motivation

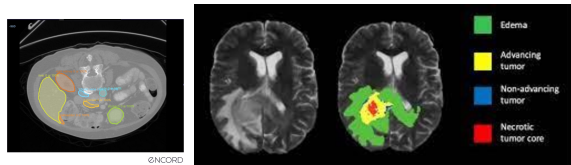
Autonomous Driving



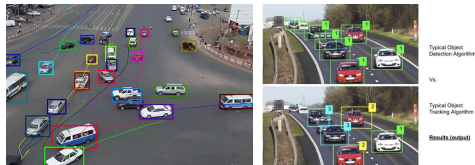
Real-World Data Complexity



Medical Imaging



Downstream Vision Tasks



Segment Anything Model (SAM)

“In this work, our goal is to build a foundation model for image segmentation.” [?]

With this model, we aim to solve a range of downstream segmentation problems on new data distributions using prompt engineering.



Problem Statement

Image segmentation is the task of grouping pixels that belong to a common semantic object and separating them from other objects and the background.

The core principle of segmentation is guided by two criteria: pixels within the same group should be as similar as possible, while pixels belonging to different groups should be as distinct as possible.

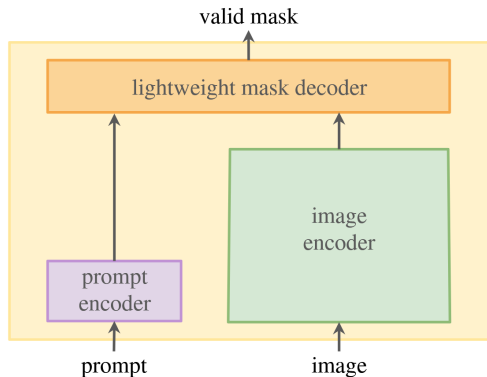
In essence, the task simulates the human cognitive ability to transform raw, unstructured visual data into a structured representation of distinct entities.



Problem Statement

For a modern promptable segmentation system like **SAM**, the problem can be formally defined by its inputs and outputs.

- **Input:** An image and a "visual prompt" that specifies the object of interest. This prompt can take various forms, such as a point (or multiple points) on the object, a bounding box enclosing it, or even a rough mask.
- **Output:** One or more segmentation masks for the object(s) indicated by the prompt, along with a confidence score for each mask. The ability to output multiple masks is crucial for handling ambiguity.



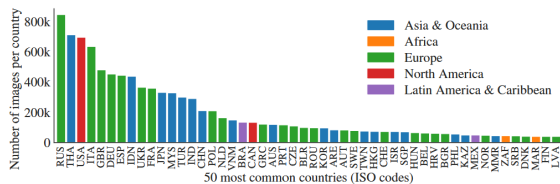
"Our principal contributions are a new task (promptable segmentation), foundational model for that new task (SAM), and dataset (SA-1B)"

A key contribution is the SA-1B dataset, which remains the largest segmentation dataset to date. Its key features include:

- **Scale:** It contains over 11 million high-resolution images and 1.1 billion high-quality segmentation masks.
- **Privacy-First:** All images are privacy-respecting, with automated blurring of faces and license plates.
- **Diversity:** The dataset is geographically diverse, with a significant number of images from countries around the world, as illustrated in Fig. 2. This helps mitigate geographical bias often present in large-scale datasets.



Contribution



The contribution of this report is to provide a comprehensive survey and structured analysis of the field of image segmentation. We synthesize the historical evolution, explain the motivations driving modern research, and offer a detailed examination of the methodology behind foundation models like SAM, contextualizing their impact and outlining future directions.

Traditional Methods

There are different techniques for image segmentation like threshold based, edgebased, cluster based, neuralnetwork based In traditional methods, there are different techniques for image segmentation like threshold-based, edge-based, cluster-based. One of the most efficient methods is the clustering method.



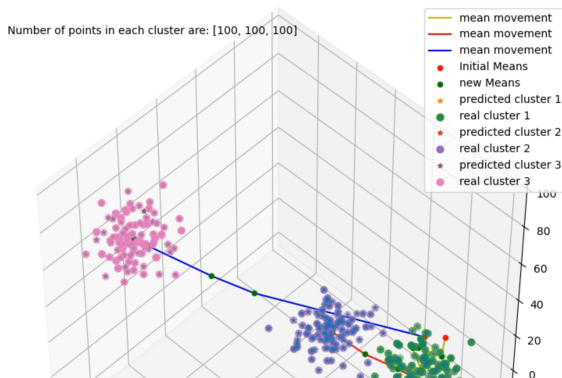
K-Means Clustering

K-Means is an unsupervised clustering algorithm used to segment regions of interest from the background [?]. It was first introduced by Hartigan in 1975, and its goal is to partition M points in an N -dimensional space into K clusters such that the within-cluster sum of squares is minimized [?].



K-Means Clustering

The algorithm iteratively assigns each pixel to the nearest cluster centroid and updates centroids based on current assignments. K-Means algorithm seeks a local optimum rather than a global one. Data points are divided into groups by assigning each of them to the nearest centroid. The goal is to find a solution such that moving any point from one cluster to another does not reduce the within-cluster sum of squares [?].



Mean Shift Clustering

Mean Shift is a non-parametric clustering algorithm, which falls into unsupervised clustering category. Unlike K-Means, it does not require specifying the number of clusters in advance. Instead, it locates regions of high data density (modes) by iteratively shifting data points toward the nearest local maximum of a kernel density estimate [?].



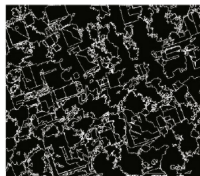
Original Satellite Imagery



Mean Shift Clustered Imagery



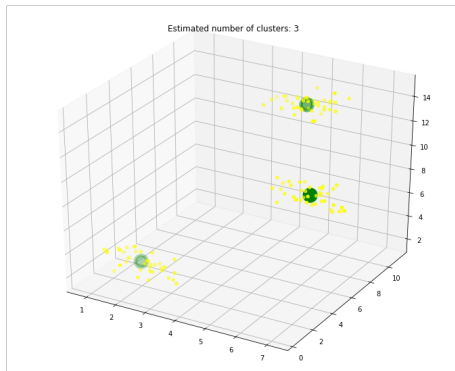
Mean Shift Labelled Imagery



Mean Shift Segmentation Boundaries

Mean Shift Clustering

The algorithm treats data points as samples from an underlying probability density function and seeks its local maxima. Each pixel is iteratively moved toward regions of higher density, determined by a kernel function, until convergence. Pixels converging to the same mode belong to the same cluster.



Binary Threshold

Binary thresholding is used to separate the foreground (region of interest) from the background based on pixel intensity values. It is particularly effective when the objects and background have distinct intensity distributions.



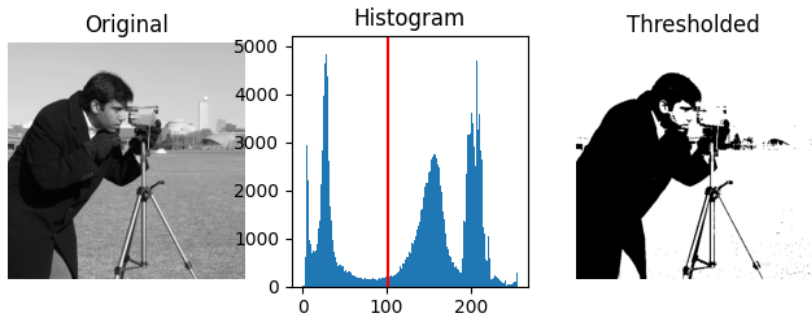
Original Image



Thresholded and segmented Image

Binary Threshold

Each pixel in the grayscale image is compared against a threshold value T . Pixels with intensity greater than or equal to T are classified as foreground, while the others are classified as background. This process converts a grayscale image into a binary one.



U-Net, proposed by Ronneberger et al. in 2015, features a symmetric encoder-decoder architecture with extensive skip connections [?]. Originally designed for biomedical image segmentation, U-Net achieves high accuracy even with limited training data.

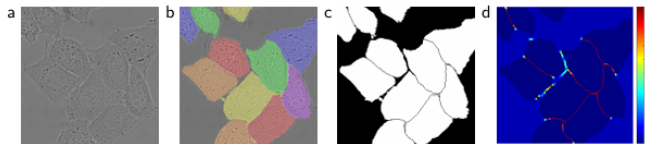
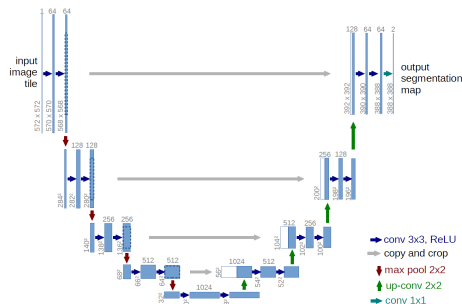


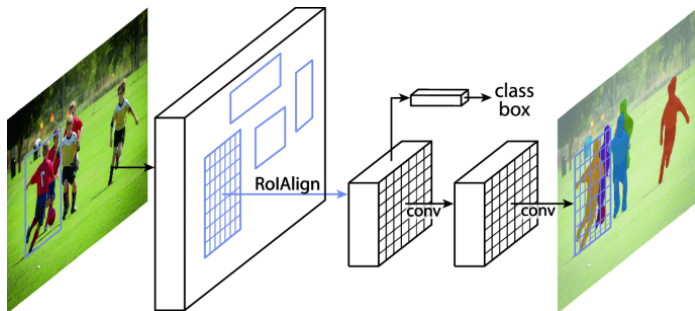
Fig. 3. HeLa cells on glass recorded with DIC (differential interference contrast) microscopy. (a) raw image. (b) overlay with ground truth segmentation. Different colors indicate different instances of the HeLa cells. (c) generated segmentation mask (white: foreground, black: background). (d) map with a pixel-wise loss weight to force the network to learn the border pixels.

FCN transforms classification networks (e.g., VGG, ResNet) into fully convolutional architectures by replacing dense layers with 1×1 convolutions. The network produces coarse feature maps through successive pooling, then upsamples them back to the original image resolution using transposed convolutions (deconvolution). Skip connections combine deep semantic features with shallow appearance features to recover spatial details lost during downsampling.



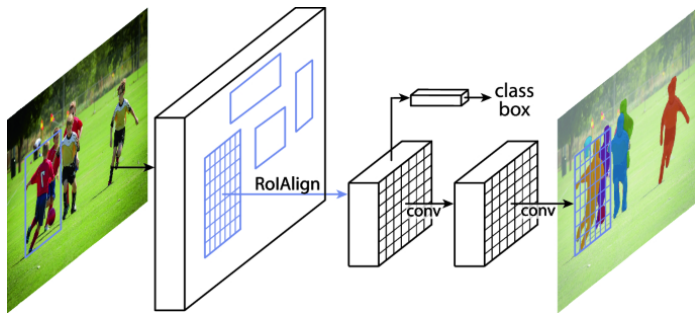
Mask R-CNN

Mask R-CNN, introduced by He et al. in 2017, extends Faster R-CNN for instance segmentation by adding a parallel branch for predicting segmentation masks [He et al., 2017]. This architecture excels at detecting and segmenting individual object instances.

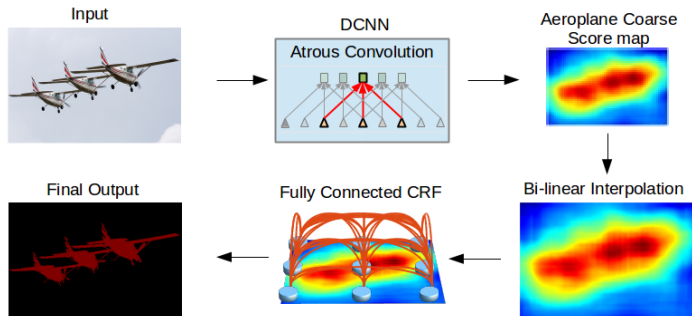


Mask R-CNN

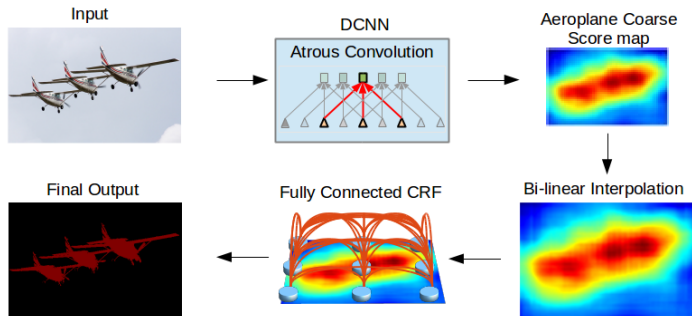
Mask R-CNN operates in two stages. First, a Region Proposal Network (RPN) generates candidate object bounding boxes. Second, for each region of interest (RoI), the network performs three parallel tasks: class prediction, bounding box regression, and mask prediction. The key innovation is RoIAlign, which preserves spatial alignment during RoI pooling by using bilinear interpolation, crucial for pixel-accurate mask prediction.



DeepLab, developed by Chen et al., employs atrous (dilated) convolution and Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale contextual information without reducing spatial resolution [?]. DeepLabv3+ further incorporates an encoder-decoder structure.

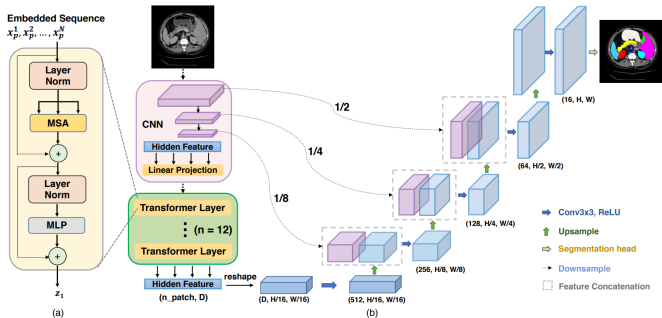


DeepLab uses atrous convolution to expand the receptive field without increasing parameters or reducing resolution. The atrous rate r controls the spacing between kernel weights. ASPP applies parallel atrous convolutions with different rates to capture objects at multiple scales. The encoder-decoder structure (in DeepLabv3+) first extracts features, then gradually recovers spatial details through the decoder with skip connections.



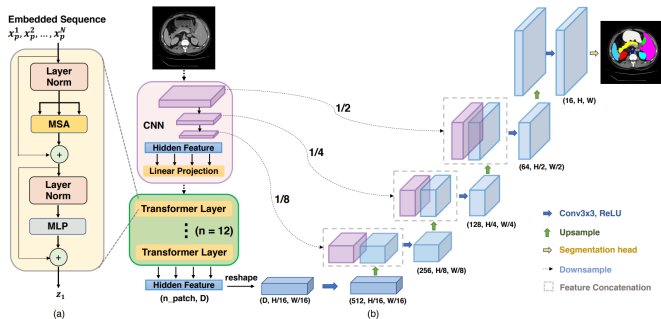
Transformer-Based Segmentation

Vision Transformers (ViTs) adapt the self-attention mechanism from NLP to computer vision. TransUNet, introduced by Chen et al. in 2021, combines Transformers with U-Net architecture for medical image segmentation [Chen et al., 2021].



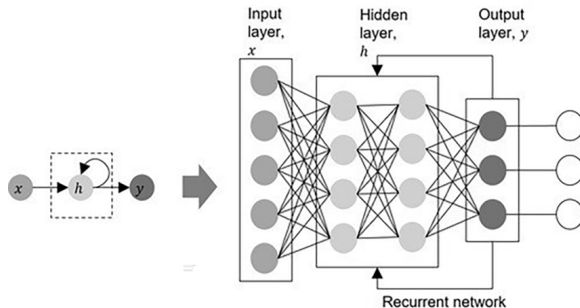
Transformer-Based Segmentation

TransUNet uses a CNN to extract initial features, then patches the feature map and processes it through Transformer encoder layers. The self-attention mechanism captures long-range dependencies between image patches, overcoming the limited receptive field of CNNs. The Transformer output is reshaped and fed into a CNN decoder similar to U-Net, with skip connections from the CNN encoder for recovering spatial details.



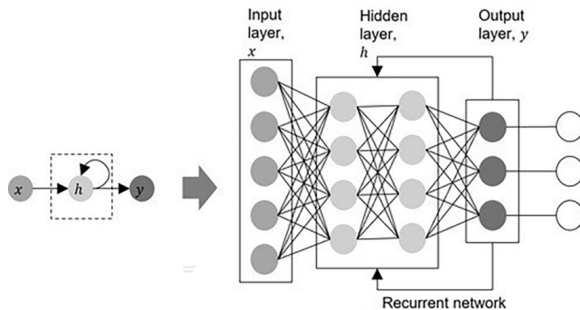
Recurrent Neural Networks for Segmentation

RNNs, particularly Long Short-Term Memory (LSTM) networks, have been applied to image segmentation to model spatial dependencies and context [?]. ReNet and related architectures process images sequentially in horizontal and vertical sweeps.



Recurrent Neural Networks for Segmentation

RNN-based segmentation treats the image as a sequence, processing it row-by-row or with overlapping patches. Four RNN layers sweep the image in different directions (left-to-right, right-to-left, top-to-bottom, bottom-to-top) to capture context from all spatial neighbors. The hidden states from all four directions are concatenated to form a rich contextual representation for each pixel. Multiple ReNet layers can be stacked to increase the receptive field.





Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A. L., and Zhou, Y. (2021).

Transunet: Transformers make strong encoders for medical image segmentation.
arXiv preprint arXiv:2102.04306.



He, K., Gkioxari, G., Dollár, P., and Girshick, R. (2017).

Mask r-cnn.

In *Proceedings of the IEEE international conference on computer vision*, pages 2961–2969.